Case Study 16: Multi-Day Water Tank Simulation

1. Problem Statement

Simulate daily water inflow and outflow for a storage tank over a multi-day period, monitoring capacity constraints, detecting spillage/overflow occurrences, and flagging critical underflow conditions when water demand exceeds available supply.

2. Objectives

- Model dynamic fluid storage levels subject to daily additions and consumption.
- Enforce upper capacity bounds and calculate overflow quantities.
- Handle deficit scenarios where demand exceeds current stored volume.

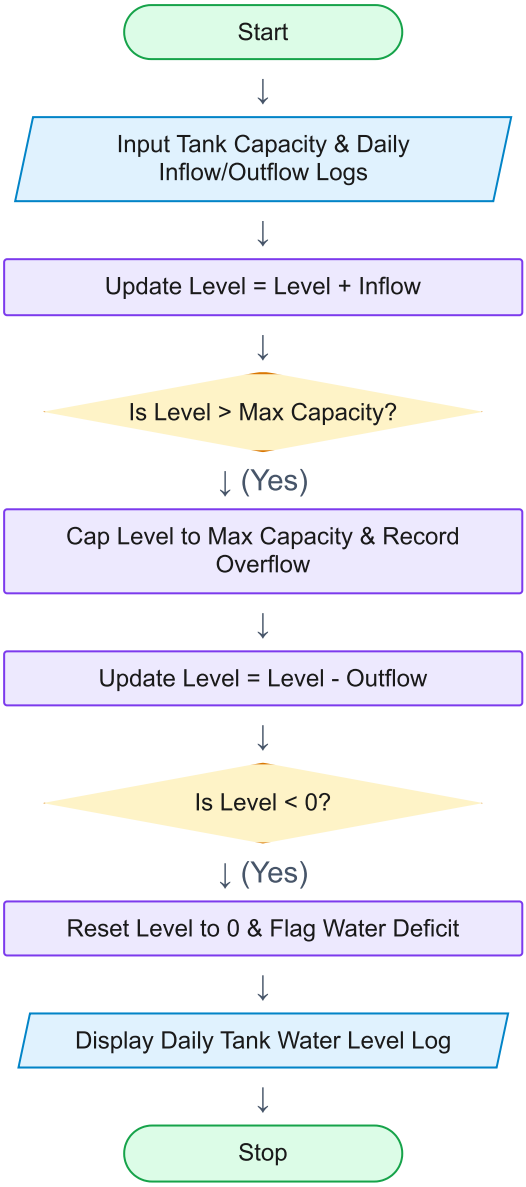
3. Assumptions

- Tank maximum capacity is fixed at **1000 Liters**.
- Daily additions occur prior to daily consumption withdrawals.
- Tank volume cannot drop below 0 Liters; any unfulfilled consumption is reported as a deficit condition.

4. Algorithm

1. Start.
2. Set tank capacity `CAPACITY = 1000` and initialize starting water level `current_level`.
3. For each day's record `(inflow, outflow)`:
   - Add `inflow` to `current_level`.
   - If `current_level > CAPACITY`:
     - Calculate `overflow = current_level - CAPACITY`.
     - Set `current_level = CAPACITY`.
   - Subtract `outflow` from `current_level`.
   - If `current_level < 0`:
     - Set deficit flag to True and reset `current_level = 0`.
   - Log daily water status, overflow, and deficit flags.
4. Display full multi-day simulation log.
5. Stop.

5. Flowchart



6. Pseudocode

```
START CAPACITY = 1000 INPUT initial_level INPUT daily_logs // List of (inflow, outflow) pairs current_level = initial_level FOR day FROM 1 TO length(daily_logs) DO inflow = daily_logs[day].inflow outflow = daily_logs[day].outflow current_level = current_level + inflow overflow = 0 IF current_level > CAPACITY THEN overflow = current_level - CAPACITY current_level = CAPACITY ENDIF current_level = current_level - outflow underflow = False IF current_level < 0 THEN underflow = True current_level = 0 ENDIF PRINT day, inflow, outflow, current_level, overflow, underflow ENDFOR STOP
```

7. Test Case and Expected Output

Simulation Parameters: Tank Capacity = 1000L, Initial Level = 300L

```
Initial Tank Level: 300 L / 1000 L ----------------------------------------------
-- Day 1: Inflow=500L, Outflow=200L -> Current Level=600L | Overflow=0L |
Deficit/Empty=False Day 2: Inflow=400L, Outflow=300L -> Current Level=700L |
Overflow=0L | Deficit/Empty=False Day 3: Inflow=100L, Outflow=700L -> Current
Level=100L | Overflow=0L | Deficit/Empty=False Day 4: Inflow=600L, Outflow=200L ->
Current Level=1000L | Overflow=300L | Deficit/Empty=False
```

8. Conclusion

The Multi-Day Water Tank Simulation accurately tracks dynamic water reserves across multi-day cycles, ensuring overflow spillage and shortage deficits are logged for resource planning.